

The Thomson Polygon from Its Cartesian Hessian: Exact Fourier Blocks and the Sharp Linear Stability Threshold

HCS Research Program

2 September 2026

Abstract

For identical positive point vortices on a planar regular N -gon, we derive the rotating relative equilibrium directly from the logarithmic Hamiltonian and reduce its raw Cartesian Hessian by a vertexwise radial–tangential discrete Fourier transform. If $c = \Gamma/(4\pi R^2)$ and $q_m = m(N - m)$, the m th Hamiltonian block is

$$c \begin{pmatrix} 0 & q_m \\ -[2(N - 1) - q_m] & 0 \end{pmatrix}.$$

Root-of-unity orthogonality proves this formula for every $N \geq 3$. Consequently the symmetry-reduced polygon is linearly elliptic for $3 \leq N \leq 6$, linearly degenerate only in the conjugate modes $m = 3, 4$ when $N = 7$, and hyperbolic for every $N \geq 8$. The heptagon statement is strictly linear. The result is a self-contained reconstruction of a classical owner theorem, not a claim of literature priority or nonlinear heptagon stability.

1 Frozen Hamiltonian and relative equilibrium

Let $z_j = (x_j, y_j) \in \mathbb{R}^2$ and give every vortex the same circulation $\Gamma > 0$. We freeze the convention

$$H(z) = -\frac{\Gamma^2}{2\pi} \sum_{j < k} \log |z_j - z_k|, \quad \Gamma \dot{z}_j = \mathbf{J} \nabla_j H, \quad \mathbf{J} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}. \quad (1)$$

Thus $-\mathbf{J}$ generates counterclockwise rotation. Take

$$a_j = R(\cos \theta_j, \sin \theta_j), \quad \theta_j = \frac{2\pi j}{N}, \quad N \geq 3, \quad R > 0. \quad (2)$$

Thomson [1] and Havelock [2] are cited as classical owners of the polygon and its linear-stability question. Cabral and Schmidt [3] provide later stability context, and Celli, Lacombe, and Pérez-Chavela [4] document polygonal relative-equilibrium lineage. No calculation or proof below is outsourced to these sources, and no literature-priority claim is made.

Lemma 1 (Angular velocity). *The configuration (2) is a relative equilibrium with*

$$\Omega = \frac{\Gamma(N - 1)}{4\pi R^2}. \quad (3)$$

For $G = H + (\Gamma\Omega/2) \sum_j |z_j|^2$, one has $\nabla G(a) = 0$.

Proof. Differentiating (1) gives

$$\nabla_j H = -\frac{\Gamma^2}{2\pi} \sum_{k \neq j} \frac{z_j - z_k}{|z_j - z_k|^2}.$$

At $j = 0$, each summand inside the sum has radial component $1/(2R)$; tangential components cancel between k and $N - k$. Rotational covariance therefore gives

$$\sum_{k \neq j} \frac{a_j - a_k}{|a_j - a_k|^2} = \frac{N - 1}{2R^2} a_j.$$

Hence $\nabla_j H(a) = -\Gamma\Omega a_j$ with (3), proving both claims and the sign of rotation under (1). $\square$

The rotating-frame linearization is

$$\mathcal{L} = \mathbf{J}_N \Gamma^{-1} D^2 G(a), \quad \mathbf{J}_N = \text{diag}(\mathbf{J}, \dots, \mathbf{J}). \quad (4)$$

All stability statements below refer to (4); they do not silently change the Hamiltonian or clock.

2 Raw Cartesian Hessian and exact Fourier blocks

For $d \neq 0$, put

$$A(d) = \frac{1}{|d|^2} - \frac{2dd^T}{|d|^4}.$$

The pairwise Cartesian blocks are

$$D_{jj}^2 H = -\frac{\Gamma^2}{2\pi} A(d_{jk}), \quad D_{jk}^2 H = +\frac{\Gamma^2}{2\pi} A(d_{jk}), \quad (5)$$

and the augmentation adds $\Gamma \Omega \mathbf{l}$ to each diagonal block. Equation (5), rather than the final closed formula, is the starting point of the independent checker.

Let $\mathbf{Q}_j$ have the radial and counterclockwise tangential unit vectors at a_j as columns. Set

$$c = \frac{\Gamma}{4\pi R^2}, \quad q_m = m(N - m), \quad \sigma_m = 2(N - 1) - q_m. \quad (6)$$

Theorem 1 (Complete DFT block). *In the local radial–tangential DFT, for every $m = 0, \dots, N - 1$,*

$$\widehat{\Gamma^{-1} D^2 G}_m = c \text{diag}(\sigma_m, q_m), \quad (7)$$

$$\mathcal{L}_m = c \begin{pmatrix} 0 & q_m \\ -\sigma_m & 0 \end{pmatrix}, \quad \mathcal{L}_m^2 = -c^2 q_m \sigma_m \mathbf{l}. \quad (8)$$

Proof. For $k \neq 0$, let $\theta = \theta_k$ and $d = a_0 - a_k$. Since $|d|^2 = 4R^2 \sin^2(\theta/2)$, substitution in $A(d)$ gives

$$A(d) = \frac{1}{4R^2 \sin^2(\theta/2)} \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}.$$

Multiplication by $\mathbf{Q}_k$ turns the displayed matrix into $\text{diag}(1, -1)$. Thus the local off-diagonal block in the first block row is

$$B_k = \frac{c}{1 - \cos \theta_k} \text{diag}(1, -1).$$

If $A_N = \sum_{k=1}^{N-1} (1 - \cos \theta_k)^{-1}$, the raw diagonal block is $B_0 = c \text{diag}(2(N - 1) - A_N, A_N)$. Pairing k with $N - k$ cancels the sine parts of the DFT and yields

$$\widehat{\Gamma^{-1} D^2 G}_m = c \text{diag}(2(N - 1) - S_m, S_m), \quad S_m = \sum_{k=1}^{N-1} \frac{1 - \cos(m\theta_k)}{1 - \cos \theta_k}. \quad (9)$$

For $\zeta = e^{2\pi i/N}$,

$$\frac{1 - \cos(m\theta)}{1 - \cos \theta} = \left| \sum_{r=0}^{m-1} e^{ir\theta} \right|^2.$$

Root-of-unity orthogonality therefore gives

$$\sum_{k=0}^{N-1} \left| \sum_{r=0}^{m-1} \zeta^{kr} \right|^2 = Nm.$$

The $k = 0$ term is m^2 , so $S_m = Nm - m^2 = q_m$. This proves (7). The local rotations preserve $\mathbf{J}$; multiplying by $\mathbf{J}$ and squaring proves (8). $\square$

3 The sharp linear threshold

Theorem 2 (Thomson polygon trichotomy). *After center, angular-impulse, and rotation reduction:*

1. every block is semisimple elliptic for $3 \leq N \leq 6$;
2. for $N = 7$, exactly $m = 3, 4$ are nonzero nilpotent blocks;
3. every $N \geq 8$ has a real hyperbolic pair.

The middle statement is linear degeneracy only; it does not assert nonlinear stability or instability of the heptagon.

Proof. For $m \neq 0$, $q_m > 0$, so (8) is elliptic, nilpotent, or hyperbolic according as σ_m is positive, zero, or negative. The maximum of q_m is $\lfloor N^2/4 \rfloor$. Hence the least sign is

$$2(N - 1) - \lfloor N^2/4 \rfloor.$$

It is positive for $N = 3, 4, 5, 6$ and zero at $N = 7$, where $m(7 - m) = 12$ only for $m = 3, 4$. For $N = 2k \geq 8$, $k^2 - (4k - 2) > 0$; for $N = 2k + 1 \geq 9$, $k(k + 1) - 4k = k(k - 3) > 0$. Thus the maximal- q_m block is hyperbolic from eight onward. The symmetry directions removed in the statement are identified explicitly below in the revised manuscript and do not alter these signs. $\square$

N	controlling σ_m	reduced linear type	strict boundary
3–6	positive	elliptic	semisimple
7	zero at $m = 3, 4$	degenerate	no nonlinear claim
8 and above	negative for some m	hyperbolic	real pair

4 Conclusion

The logarithmic pair Hessian and one root-of-unity sum determine every Fourier block of the equal-vortex polygon. Once rotation, scale, and translations are handled explicitly, the sign $2(N - 1) - m(N - m)$ yields the exact reduced linear transition at seven and eight vortices.

References

- [1] J. J. Thomson, *A Treatise on the Motion of Vortex Rings*, Macmillan, London (1883).
- [2] T. H. Havelock, “The stability of motion of rectilinear vortices in ring formation,” *Philosophical Magazine* **11** (70), 617–633 (1931), doi:10.1080/14786443109461714.
- [3] H. E. Cabral and D. S. Schmidt, “Stability of relative equilibria in the problem of $N + 1$ vortices,” *SIAM J. Math. Anal.* **31** (2), 231–250 (2000), doi:10.1137/S0036141098302124.
- [4] M. Celli, E. A. Lacomba, and E. Pérez-Chavela, “On polygonal relative equilibria in the N -vortex problem,” *J. Math. Phys.* **52**, 103101 (2011), doi:10.1063/1.3646115.